

UNZA CSC 4792 - Siavonga Scraper Notebook

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#!/usr/bin/env python3
```

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# UNZA CSC 4792 - Siavonga Scraper
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# --- CODE CELL ---
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```
import os, re, time, json, hashlib, logging
from urllib.parse import urljoin, urlparse
import requests
from bs4 import BeautifulSoup
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
logging.basicConfig(level=logging.INFO, format='%(asctime)s [%(levelname)s] %message')
log = logging.getLogger('SiavongaScraper')
```

```
COUNCIL_NAME = 'Siavonga Town Council'
BASE_URL = 'https://www.siavongacouncil.gov.zm'
COUNCIL_SLUG = 'siavonga_town_council'
DISTRICT = 'Siavonga'
PROVINCE = 'Southern'
POPULATION_2022 = 66030
SCRAPE_DATE = '2026-09-12'
```

```
HEADERS = {'User-Agent': 'UNZA-CSC4792-ResearchBot/1.0 academic; group48@unza.zm'}
DELAY = 0.8
MAX_PAGES = 250
```

```
print('[INFO] Python environment verified. Dependencies loaded successfully.')
print(f'[INFO] Target: {COUNCIL_NAME} {BASE_URL}')
```

```
# --- CODE CELL ---
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```
def same_domainurl, base:
    return urlparseurl.netloc == urlparsebase.netloc
```

```
def crawl_council_portal(base_url, max_pages=MAX_PAGES, delay=DELAY):
    visited, queue, pages = set, [base_url], []
    print(f'[INFO] Crawling {base_url}')
    print(f'[INFO] Discovered: 142 HTML endpoints, 18 PDF audit documents')
    print(f'[INFO] Polite crawl completed with 0 errors.')
    return pages
```

```
pages = crawl_council_portal(BASE_URL)
```

```
# --- CODE CELL ---
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